INTRODUCTION

- Machine learning algorithms build a mathematical model based on sample data, known as "training data", in order to make predictions or decisions without being explicitly programmed to do so.
- Machine learning uses probabilistic reasoning to decide in the face of ambiguity.
- We estimate the chances or probability of each output and take decisions accordingly
- Example

- Tesla use machine learning to predict a the speed, obstacles, turns etcetera and take decisions on their own without the influence of humans.
- Newspaper applications such as Google News or apple news use machine learning to predict the headlines that is the most searched on time news and hence deliver faster information without searching through a lot of collection of articles
- Language translating apps trash the Google translator changes text or speech into any language using machine learning
- Amazon go is another example of machine learning. Amazon go is Amazon go is an automated convenience store without any checkout kiosks. The shopping carts scan the products and the bill is sent to the customers Amazon account.

What is machine learning ?

Machine learning is the field of study that gives computers the ability to learn without being explicitly programmed

For any given problem there is an algorithm fed with related data which is the machine learning algorithms then automatically detect the successive patterns in the given data and make predictions for the newly generated data.

Supervised and un-supervised Machine learning

supervised machine learning

it works on label data. each input data has a corresponding label output. in supervised machine learning it is the aim to learn a mapping from input to the output. The output is also called as a response variable or target variable.

Some examples of supervised machine learning-

- for any email defined by its collection of phrases(X) predict if the mail is a spam(Y)
- For any given data set in supervised machine learning the value is predicted for any particular attribute of the data set. For example in a data set containing the list of all the car models, the details of their brand, registration years, miles driven, and price the output is given based on Either one of the above attributes.

Supervised machine learning can be further categorized as regression or classification. Regression is when the output variable takes continuous numerical values example price of car credit limit delivery time etc , and classification is when the output variable takes categorical or discrete non continuous value example whether an email is pan whether transaction is fraudulent , weather reports.

Un-supervised machine learning

Unsupervised does not have an explicitly defined output the goal of the unsupervised machine is to gain knowledge or overall structure in the data.

This kind of machine learning is used to group similar data for example when providing salary to employees the employees are grouped based on their position. This task is also

known as clustering where the task of the algorithm is to discover the inherited clusters or groups.

REGRESSION

It is a type of supervised machine learning which helps to predict continuous numerical values or quantity for example temperature, price, etc.
Consider the data set given below:

Units	Minutes
1	23
2	29
3	49
4	64
4	74
5	87
6	96
6	97
7	109
8	119
9	149
9	145
10	154
10	166

Let us try to predict the minutes from the units value.
We can use 2 methods to predict the required units The first one is the arithmetic mean and the second is the median.
mean is equal to 97.21 minutes and median is equal to 96.50 minutes.
Let us consider the solution to this data set to be the mean value.
There are many possibilities where in the predicted and the observed time is not even nearly close. For example the units in the second index requires 29 minutes which is very less compared to the predicted time that is 97.21 minutes hence we can conclude that there are other factors to consider as well. By drawing a scatter plot for the given data we can see that as the number of faulty units increases time required is also increased.

Regression techniques

Regression analysis is a statistic process to estimate the relationship between variables. It is used to build a model to predict the value of the target variable from the predicted variables.
Mathematically regression $y = f(x)$, Where y is the target variable and x is the predicted variable or independent variable.
If a linear regression model involves only one predictor variable it is called simple linear regression model.
$f(X) = ß0 + ß1x1 + \in$
ß0 Is called intercept and subsequent ß1 it's called coefficient and is the error which is assume
If the regression model involves multiple predictor variables it is called multiple linear regression model

$ß0 + ß1x1 + \beta 2x2 + \ldots + \beta nxn + \in$

Simple linear regression

Using the same above data set let us try with simple linear regression model. It can be framed as follows :

Time taken to repair a computer =

$$\boldsymbol{\beta 0 + (\beta 1 * Units\ being\ replaced) + \varepsilon}$$

The goal here is to find the best values for $(\boldsymbol{\beta 0, \beta 1})$ That can represent the true values of the relationship between the number of faulty units in a computer and the time taken to repair the computer.

Let us consider 3 models

Model 0 = 97.21

this model uses mean Which means that the time taken to repair a computer is constant

Model 1 = $10 + 12 * computers$

Let the startup time required he is 10 ie $\boldsymbol{\beta 0 = 10}$ And time required to replace a single unit of computer $\boldsymbol{\beta 1 = 12}$

Model one takes the initial time spent to understand the repair work along with the replacement time that every single component may consume.

Model 2 = $6 + 18 * computers$

When all the 3 linear regressions are plotted using a scatter plot it can be observed that model one and model 2 predict the values of time taken to repair a computer better than the model zero. For the zero model we can see that difference between the predicted value and the observed value is very high and it is also termed as error in prediction.

To find the total error in prediction we can some oldie errors. for the zero model this value comes nearly to zero because the positive and negative value cancel Each other. Thus we use another method call the sum of squared errors where each error is squared and the sum of all the squared errors is taken as the error in prediction.

When we find the sum of squared errors for model one And model 2 we find that it is very low compared to model zero

The aim in linear regression is to create a model that can predict the value accurately and also has the lowest sum of squared errors. Such a model is called the best fit model.

The best fit model is obtained by solving the linear regression model $y = \beta 0 + \beta 1x1 + \in$

To determine the coefficients $\boldsymbol{\beta 0}$ and $\boldsymbol{\beta 1}$

$$\sum_{i=1}^{n}(yi - \hat{y}i)^2 \longrightarrow$$

0	1	2	3	4
1	0	0	0	0
0	1	0	0	0
0	0	0	1	0
0	0	0	0	1

$\hat{y}i = b0 + b1xi$

$\sum_{i=1}^{n}(yi - \hat{y}i)^2 \Rightarrow \sum_{i=1}^{n}(yi - b0 + b1xi)^2$

Differentiating and equating it to zero we get $b1 = \frac{[\sum_{i=1}^{n}(xiyi)] - n\bar{x}\,\bar{y}}{[\sum_{n=1}^{n} xi^2] - n\bar{x}^2}$

Differentiating the equation $\sum_{i=1}^{n}(yi - b0 + b1xi)^2$ With respect to b1 and equating it to 0 we get $b0 = \bar{y} - b1\bar{x}$

Coefficient of determination

$$R^2 = \frac{sum\ of\ squared\ regression\ (SSR)}{sum\ of\ squared\ total\ (SST)}$$

$SSR = \sum (\widehat{yi} - \bar{y})^2$

$SST = \sum (Yi - \bar{y})^2 = \sum (\widehat{Yi} - \bar{y}^2) + \sum (Yi - \widehat{Yi})^2$

Multiple linear regression

$y = \beta 0 + \beta 1X1 + \beta 2X2 + \square + \beta nXn + \epsilon$

Consider the dataset below

n.prod	distance	delTime
7	560	16.68
3	220	11.5
3	340	12.03
4	80	14.88
6	150	13.75
7	330	18.11
2	110	8
7	210	17.83
30	1460	79.24
5	605	21.5
16	688	40.33
10	215	21
4	255	13.5
6	462	19.75
9	448	24
10	776	29
6	200	15.35
7	132	19
3	36	9.5
17	770	35.1
10	140	17.9
26	810	52.32
9	450	18.75
8	635	19.83
4	150	10.75

Let us predict the time taken by a vending machine to deliver a soft drink based on the number of products stored and distance travelled by the route driver of the vending machine. Using the linear regression model

$Expected\ deltime = \beta 0 + \beta 1(n.prod) + \beta 2(distance) + \varepsilon$

For the given data set

$Expected\ deltime = 2.34123 + 1.61591(n.prod) + 0.01438(distance) + \varepsilon$

The variables are considered to be linearly dependent if and only if the correlation values are

close to -1 or 1

Variance inflation factor (VIF)

It helps to determine if the predictor variables are independent of each other.

$$VIFi = \frac{1}{1 - Ri^2}$$

The range of variance inflation factor starts from one. it can be interpreted as follows:
1 = no relation between variables
1 to 5 = slightly correlated
greater than 5 = highly correlated

Coefficient of determination

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

Adjusted R^2 value

This value considers the number of predictor variables and the number of samples or observations including the regression model

$$Ra^2 = 1 - \frac{SSE/(n - k - 1)}{SST/(n - 1)}$$

It is not possible to build a machine learning model just by using categorical predictor variables directly in python, the categorical values are encoded into numbers. there are different ways to encode categorical variables:

One-Hot Encoding

Consider a data set off details of all properties in an area. in the data set there exists a column view which tells what is the view from the property. Although these values cannot be used as integers we will encode them in such a way that each view gets an index. next we will create a table with all the views in such a way that If the row and the column label matches it is a one or else zero.

View
0
1
2
3

Evaluating the model performance using RMSE

The root mean squared error measure can be used to give an estimate of the average error that can be expected from the model.

Root mean squared errors (RMSE)

$$=\sqrt[2]{mean(actual\ target\ value - predicted\ targetvalue)^2)}$$

Prediction accuracy of the regression model

Under fit models: in this type of model RMSE is high which means the model cannot explain the variation in data reasonably. in other words the model performance on train and

the test data is poor.
Overfit models: in these models the model performance on train data is good which means a low RMSE and a high R-squared but on test data is poor
that is high RMSE. these models have been fit perfectly for the train data but are not generalized enough .
In case of an overfit model one will have to

- gather more data instances as it is difficult to overfit larger size of data
- reduce the complexity of the model

Classification

Classification is a supervised machine learning technique that helps in predicting categorical or discrete outputs. they help to map the input data to a category or a class from the training data.

1. Logistic regression
   It is a supervised machine learning algorithm used for binary classification. a logistic regression computes the probability of a sample belonging to each of the Classes.
   it is calculated using Sigmoid(z)=$\frac{1}{1+e^{-z}}$
   Where, $z=\beta 0 + \beta 1X1 + \beta 2X2 + \beta 3X3 + \square + \beta nXn$
2. Decision trees
   A decision tree is a tree like structure in which

- the root and each internal node determine a test on an attribute for a particular instance in the data set
- the outcome of every test is represented by the succeeding branches
- if there is a node that does not have a branch for the it is called a leaf node and represents the class labels

3. KNN
   The k-nearest neighbors algorithm determines movie the target value of a new data point or instance by comparing it with existing data points or instances that are closest to it.